

INC-001: Missing Sales Data Incident RCA

Incident Report: INC-001 - FactSales Missing 3 Days of Data

Incident ID: INC-001
Severity: P1 - Critical
Status: Resolved
Reported by: Chloe Tan (BI Developer)
Date reported: 2026-10-15 08:47 UTC
Date resolved: 2026-10-15 14:22 UTC
Duration: 5 hours 35 minutes
Affected tables: marts.fact_sales
Missing data: 2026-10-12, 2026-10-13, 2026-10-14 (3 days)

1. Business Impact

The Power BI trading dashboard showed no sales data for 12-14 October 2026.
Chloe Tan reported this to the team after the trading director asked for a daily revenue figure at the 09:00 standup.
Finance could not produce the daily revenue report.
Revenue at risk of misreporting: approximately £1.2M across 3 days (estimated from prior week average).

2. Timeline

Time (UTC)	Event
2026-10-12 06:45	Airflow DAG northpeak_daily_sales shows SUCCESS but dbt_test_gold_models task failed silently (exit code 1 not propagated)
2026-10-13 06:40	Same failure repeats - not noticed
2026-10-14 06:38	Same failure repeats - not noticed
2026-10-15 08:47	Chloe Tan reports missing data to engineering team
2026-10-15 08:55	On-call engineer (YOU) picks up incident
2026-10-15 09:10	Investigation starts: Airflow logs reviewed
2026-10-15 09:35	Root cause identified: dbt test failure not propagating exit code to Airflow
2026-10-15 10:00	Fix deployed to DAG (trigger_rule changed, exit code handling fixed)
2026-10-15 10:15	Backfill triggered for 3 missing dates
2026-10-15 14:22	Backfill complete, data validated, incident resolved

3. Root Cause Analysis

Root cause: The `dbt_test_gold_models` BashOperator was using `bash_command` without `pipefail`, meaning a non-zero exit code from `dbt test` was swallowed by the shell. Airflow marked the task as `SUCCESS` because the `bash` wrapper returned 0.

Why not detected immediately: The monitoring dashboard only checked DAG run status (`SUCCESS/FAILED`), not task-level exit codes. A green DAG run was treated as data-available.

Contributing factor: No row count validation after `dbt` run. If we had checked that `fact_sales` received at least N rows for the execution date, the failure would have been caught within minutes of the 06:00 run completing.

Wrong direction trap: Initial investigation suspected a source system outage (NP Core). The PostgreSQL connection logs were clean. Wasted 20 minutes before pivoting to the Airflow task logs.

4. Resolution

1. Fixed BashOperator `bash_command` to use `set -e` and `set -o pipefail`
2. Added `on_failure_callback` to send PagerDuty alert on any task failure
3. Added post-dbt row count validation task to the DAG
4. Backfilled 3 missing dates using `airflow dags backfill`

5. Prevention Actions

Action	Owner	Due Date	Status
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Add row count validation task after all dbt runs	Data Platform Team	2026-10-22	Done
Add PagerDuty integration to all P1-severity DAG failures	Data Platform Team	2026-10-22	Done
Update monitoring dashboard to show task-level status	Data Platform Team	2026-10-29	In Progress
Add Great Expectations freshness check post-dbt	Data Platform Team	2026-10-29	To Do
Document backfill runbook step in production_runbook.md	Data Platform Team	2026-10-29	To Do

6. Lessons Learned

- **Never trust DAG-level SUCCESS alone.** Always validate data presence, not just execution status.
- **BashOperator exit code handling is a known footgun.** Use `set -e -o pipefail` in every `bash_command`.
- **Monitoring must check data, not process.** A pipeline that runs but produces no data is a failure even if the DAG is green.
- **The wrong direction trap in investigation:** always check Airflow task logs before assuming source system failure.